

Article

Automatic Vehicle Accident Detection from Images During Adverse Weather Conditions Using Yolo10 and VGG19

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Abstract: This study addresses the challenges of detecting vehicle accidents in adverse weather conditions, where poor visibility, environmental noise, and complex scenes hinder accurate detection. Given the increasing importance of road safety, particularly for autonomous and assisted driving systems, reliable accident detection is essential. This research proposes a hybrid approach that combines deep learning models, specifically, YOLOv10 and VGG19, to enhance detection accuracy and robustness in images captured under adverse conditions. A dataset of 3,480 annotated images was used, with specific subsets allocated for each model: YOLOv10 was trained on 1,307 images (train: 915, validation: 261, test: 131), while VGG19 utilized 1,679 images (train: 1,175, validation: 336, test: 168). The VGG19 test set was further categorized into 90 accidents and 78 non-accident images, with similar categorization in training (596 accident, 579 non-accident) and validation (169 accident, 167 non-accident) subsets. YOLOv10, known for its real-time object detection capabilities, achieved precision values of 54.2%, 63.2%, and 81.8% over epochs 50, 100, and 200, respectively, demonstrating strong adaptability to challenging conditions like rain, fog, snow, and low light. The VGG19 model with a fully connected layer showed the highest accuracy, precision, recall, and F1-score, particularly at epoch 200, excelling in detecting subtle indicators such as minor vehicle damage or unusual object interactions. This dual model framework outperforms traditional single-model approaches, leveraging YOLOv10's speed and efficiency alongside VGG19's depth in feature extraction. Experimental results confirm that this combined approach significantly reduces false positives and improves detection accuracy, this study underscores the potential of hybrid deep learning models for advancing accident detection systems, contributing to intelligent transportation by providing a reliable tool for real-time accident identification under adverse weather conditions. This approach marks an important step in traffic safety, supporting faster emergency response and enhancing overall road safety.

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1. Introduction

Ensuring road safety in adverse weather conditions is crucial, as factors such as poor visibility, altered road friction, and reduced maneuverability in rain, fog, snow, and low light significantly increase the risks of accidents and fatalities [27]. Traffic injuries remain one of the leading causes of death globally, underscoring the need for reliable and timely accident detection systems, particularly in challenging weather conditions [28]. To address this, recent research has focused on developing more accurate and responsive detection methods that can operate in real-time and adapt to varying environmental factors [29].

Traditional accident detection approaches, which include manual monitoring and basic sensor-based systems, often prove inadequate in low-visibility scenarios due to environmental constraints, emphasizing the need for advanced automated detection techniques [30]. Recent advancements in computer vision and deep learning have demonstrated potential in overcoming these limitations by leveraging image analysis to improve both the speed and accuracy of detection [31]. Notably, models such as YOLO (You Only Look Once) and VGG (Visual Geometry Group) have shown significant efficacy in object detection and scene analysis, laying a strong foundation for sophisticated real-time road monitoring systems.

Deep learning applications in accident detection typically involve training neural networks to recognize patterns indicative of collisions or hazardous situations in visual data. YOLO models, for example, are designed for real-time object detection, processing images in a single pass, making them highly suitable for high-speed applications such as traffic surveillance. The latest iteration, YOLOv10, introduces architectural enhancements that enable it to handle complex scenes effectively, even under adverse weather conditions. Meanwhile, the VGG19 model, a deep convolutional neural network, is known for its capability to extract detailed visual features, which allows it to detect subtle indicators of accidents, such as minor vehicle damage or abnormal positioning, essential for accurate accident identification. However, integrating these models to optimize detection accuracy poses a technical challenge, especially in low-visibility environments where reduced visibility and image noise can result in false positives or missed detections.

To address these challenges, this study proposes a dual-model accident detection framework that combines YOLOv10's real-time detection strengths with VGG19's detailed feature extraction, further enhanced by adaptive image preprocessing techniques. Techniques such as contrast enhancement and image augmentation are applied to improve image clarity, providing more accurate data for both models and reducing detection errors in adverse weather conditions. This dual-model fusion approach represents a shift from traditional single-model strategies, merging the speed of YOLOv10 with the analytical precision of VGG19 to create a more robust detection system.

The primary objective of this study is to develop a reliable, automated vehicle accident detection system tailored for adverse weather conditions, supporting real-time applications across diverse traffic environments. By integrating adaptive preprocessing strategies with a complementary model fusion approach, this research aims to enhance accident detection capabilities under challenging weather, contributing to safer and more responsive transportation networks globally. This work aligns with the broader goal of improving public safety and reducing road fatalities through innovative, technology-driven solutions.

In summary, this study builds upon previous research that underscores the importance of multi-model approaches in accident detection, particularly in adverse weather scenarios. While prior studies have explored the potential of single models or static detection methods, this framework

introduces an adaptive, dual-model approach that merges real-time detection with comprehensive feature analysis, offering a more accurate and thorough accident detection system.

2. Related Work

Various studies have investigated deep learning and sensor-based frameworks to enhance accident detection, especially under challenging weather conditions. This section reviews notable works that leverage advanced models, multimodal data, and preprocessing techniques to improve the accuracy and robustness of accident detection systems in adverse environments.

Kim et al. [1] developed a smart road framework that leverages roadside sensors and communication networks to detect accidents in real time, especially during poor weather. By monitoring road conditions through sensors rather than in-vehicle systems, this framework achieved cost-effectiveness and efficiency in accident detection. Additionally, it enabled real-time alerts for approaching vehicles, thereby reducing multi-vehicle collision risks under adverse conditions.

Simonyan and Zisserman [3] introduced the VGG19 model, a deep convolutional neural network consisting of 19 layers, which excels at hierarchical feature extraction from images. VGG19 was applied to accident detection tasks, where it demonstrated high accuracy in identifying accident indicators such as minor vehicle damage or unusual vehicle positioning under low visibility conditions, including rain, fog, and snow. This capability makes VGG19 a valuable tool for understanding complex road scenes.

Rahnemoonfar and Shetty [5] evaluated YOLO, a deep learning model known for real-time object detection, across varied weather conditions. They trained YOLO on synthetic datasets simulating extreme weather, achieving high detection accuracy even in challenging scenarios. The model's adaptability to diverse conditions suggests its suitability for applications requiring fast response times, such as real-time accident detection during severe weather.

Zhou et al. [4] combined YOLO and VGG architectures to create a weather-adaptive deep learning model. By integrating preprocessing methods such as dehazing, contrast enhancement, and noise reduction, they dynamically adapted detection models to different weather scenarios. This approach significantly enhanced detection accuracy and minimized errors, underscoring the importance of weather-specific adaptations for reliable accident detection.

Heinzler et al. [6] and Hnewa and Radha [7] focused on the limitations of conventional computer vision algorithms for object detection in adverse weather, emphasizing the need for advanced models like YOLO and VGG. They also advocated for the use of multi-modal sensor data (e.g., cameras, LiDAR, radar) to boost detection reliability, supporting the robustness of accident detection in low-visibility conditions.

Han et al. [2] utilized CNNs and LSTM networks to detect and track vehicles in adverse weather. Training these models on augmented datasets simulating fog, rain, and snow significantly improved detection accuracy. This study demonstrated the effectiveness of deep learning in traffic monitoring under challenging conditions.

Du et al. [9] discussed the use of image enhancement techniques, such as dehazing and contrast adjustment, to improve the input quality for accident detection models. By ensuring YOLOv10 and VGG19 operate on high-quality data, they reduced error rates and improved detection accuracy, highlighting the importance of preprocessing in adverse weather scenarios.

Gupta and Singh [10] proposed a multi-modal data fusion approach that integrates sensor data with deep learning models for vehicle detection in foggy conditions. Their method combined YOLO's real-time detection with VGG's feature extraction, enhancing the system's understanding of traffic scenes and demonstrating the potential of multi-modal approaches for low-visibility environments.

Chen et al. [11] enhanced YOLO and VGG models for low-visibility conditions, such as fog and rain. The modified models maintained real-time performance while improving detection accuracy, making them well-suited for accident detection in adverse weather. This study illustrates the benefits of model adaptations for specific visibility challenges.

Lee et al. [12] examined how inclement weather impacts road surface conditions and accident rates, showing that rain, snow, and ice significantly increase crash occurrences. Their findings underscore the value of integrating real-time road monitoring with accident detection systems to enhance traffic management during adverse weather.

Jiang et al. [8] explored the use of synthetic training data and transfer learning to increase model robustness. Training YOLO and VGG on datasets simulating fog, rain, and snow improved detection accuracy and reliability. Transfer learning further enabled these models to adapt efficiently to new environments, making them effective for real-world accident detection scenarios.

To conclude, the reviewed studies highlight different techniques for accident detection, including image enhancement and sensor fusion, but many rely on custom or synthetic datasets to handle weather challenges. However, there is a noticeable gap in research specifically focused on detecting accidents in bad weather. Our project aims to address this by developing a system that combines accident detection with real-time weather adjustments, offering a practical approach to improve road safety under adverse conditions.

3. Materials and Methods

This study focuses on vehicle accident detection under adverse weather conditions using deep learning models. The methodology involved several key steps: First, a dataset was gathered for training, validation, and testing, with approximately 70% of the images allocated for training, 20% for validation, and 10% for testing. Images were collected from online platforms and supplemented with frames extracted from short videos using the Roboflow platform, which also facilitated annotation for identifying vehicles and accident indicators. Next, the YOLOv10 model was trained for real-time object detection, while VGG19 was used for detailed feature extraction. Both models were trained and evaluated on Google Colab, with performance measured using accuracy and precision metrics. This structured approach enabled the development of a robust system for vehicle accident detection.

3.1. Data Gathering

Since no pre-existing dataset met our requirements, we gathered the dataset specifically for this study. A total of 1,307 images were collected for the YOLO model, with 915 images allocated for training, 261 for validation, and 131 for testing. For the VGG model, a dataset of 1,679 images was compiled, divided as follows: 1,175 images for training (596 accident, 579 no accident), 336 for validation (169 accident, 167 no accident), and 168 for testing (90 accident, 78 no accident). All images were sourced from YouTube, and video frames were extracted and annotated using the Roboflow platform.

3.2. Dataset Splitting

The final dataset was divided into training, testing, and validation sets. For both the YOLO and VGG models, the training set comprised 70% of the total data, while 20% was allocated for validation, and the remaining 10% was used for testing. To ensure the model's ability to generalize and handle diverse data, data augmentation techniques were applied to the validation set. This approach helps the model adapt to various real-world conditions and improves its robustness in recognizing vehicle accidents in different weather scenarios. By carefully splitting the dataset and applying augmentation, we aimed to maximize the model's performance and accuracy across all stages of training, validation, and testing.

3.3. Neural Network Algorithms

Selecting the right algorithms is essential in computer vision, as the model choice significantly influences detection accuracy and speed. For detecting vehicle accidents under challenging conditions, this study prioritized models known for high accuracy and robust feature extraction: YOLO, CNN, and VGG19. The YOLO algorithm is widely recognized for its efficiency in real-time object detection, enabling rapid identification of potential accident indicators. CNNs and VGG architectures, on the other hand, excel in capturing complex visual features through deep, hierarchical

layers, allowing the models to progressively learn detailed accident-related patterns. This combination of YOLO's speed with the depth of CNN and VGG19 makes these algorithms well-suited for detecting subtle indicators of accidents, even under adverse weather conditions.

3.3.1. The YOLO Model

YOLO, which stands for “You Only Look Once,” is a state-of-the-art object detection model known for its ability to perform real-time object detection with high accuracy.

YOLO processes an entire image in a single pass, making it suitable for applications that demand both speed and accuracy, such as accident detection in dynamic environments. YOLO's unique architecture divides the input image into a grid and predicts bounding boxes and class probabilities for each cell, enabling the model to identify multiple objects simultaneously, as shown in Fig. 1.

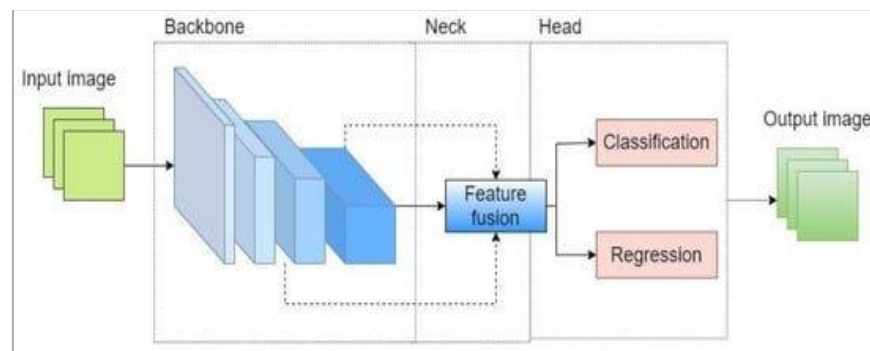


Figure 1. The architecture of the YOLO model, inspired by [20].

In this study, we implemented YOLOv10 to take advantage of its advanced capabilities in real-time detection, including Non-Maximum Suppression (NMS) to reduce redundant bounding boxes and ensure precise object localization. YOLOv10 was specifically chosen for its rapid response time, allowing it to detect vehicles, pedestrians, and road elements under challenging conditions, including rain, fog, and low light. This fast and efficient model serves as the primary layer for real-time accident detection, enabling quick identification of potential hazards in adverse weather conditions.

3.3.2. The CNN Model

Convolutional Neural Networks (CNNs) are widely recognized for their robust image processing capabilities. They are particularly suited for tasks involving image classification and feature extraction, making them a foundational architecture for accident detection systems. CNN has four layers: convolution layer, pooling layer, fully connected layer, and nonlinearity layer [17]. Illustrations of those four layers are presented in Fig. 2. Further CNNs function by applying a series of convolutional filters over an input image, allowing the model to capture spatial hierarchies in visual data. This layered approach makes CNNs highly effective at identifying patterns within complex scenes, such as accident-prone scenarios during adverse weather conditions.

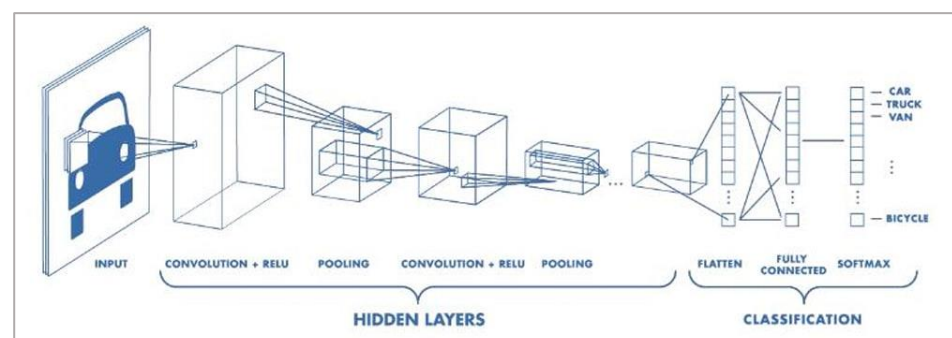


Figure 2. The architecture of the CNN model, inspired by [18].

In this study, CNN served as a preliminary model, providing an essential basis for understanding key visual patterns associated with vehicle accidents. By using straightforward CNN architecture, we aimed to capture fundamental accident indicators, such as the presence of vehicles and road obstacles. Although CNNs are effective in initial image processing, their detection capabilities are further enhanced when combined with more specialized models like YOLO and VGG, which offer additional speed and detail for accident detection.

3.3.3. The VGG19 Model

The VGG model, developed by the Visual Geometry Group, is renowned for its deep convolutional architecture, with VGG19 as shown in Fig. 3 it is containing 16 convolutional layers and 3 fully connected layers. VGG excels at extracting fine-grained features from images, which makes it highly effective in recognizing subtle accident indicators that may go unnoticed in real-time detection models. The depth of the VGG19 architecture allows for detailed hierarchical feature extraction, enabling the model to capture intricate image details that are essential for accurate accident detection in complex environments.

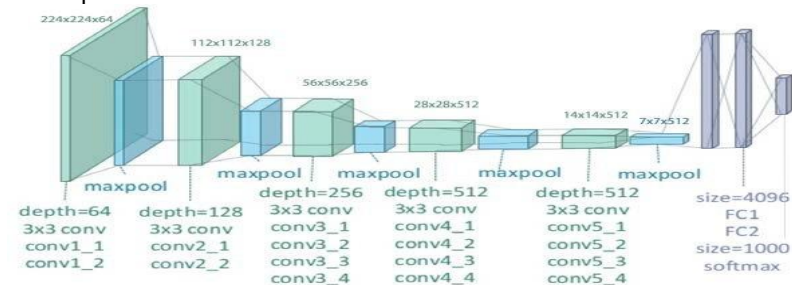


Figure 3. The architecture of the VGG19 model, inspired by [19].

In this research, VGG19 complements YOLO by providing an additional layer of feature extraction on the regions identified by YOLOv10. This dual-model approach leverages the strengths of both models: YOLOv10's speed for initial detection and VGG19's depth for detailed analysis, resulting in improved detection accuracy and reduced false positives. This combination ensures a robust and reliable detection system capable of identifying accident-related cues even under adverse weather conditions.

3.4. Build and Train the Models

The model-building phase was conducted on Google Colab, which provided the necessary computing power and GPU resources to train deep learning models efficiently.

First, we implemented the YOLOv10 model, leveraging its pre-built architecture for real-time object detection. Using Google Colab, we set up the YOLOv10 environment, loaded the pre-trained weights, and adjusted only the batch size and number of epochs to optimize detection of vehicle accident indicators.

For the CNN model, we built an architecture tailored to classify images into accident and non-accident categories. We experimented with different configurations, including a 7-layer CNN without dropout, a 7-layer CNN with dropout to reduce overfitting, and a deeper 10-layer CNN without dropout layers to enhance feature extraction. For VGG19, we trained both a version with the fully connected layer and one without it, to analyze their effects on model accuracy, with adjustments to learning rate, batch size, and epochs to refine accuracy and minimize loss.

3.5. Training Hyperparameters

To optimize the performance of the accident detection model, a series of adjustments were made to the training parameters, as outlined in Table 1. These adjustments primarily focused on the number of epochs, batch size, and learning rate. Initially, we experimented with different epoch values, selecting 50, 100, and 200 epochs to evaluate their impact on model convergence and training stability. Additionally, the batch size was set to 32 for all experiments to balance the efficiency and performance during training. For the learning rate, we utilized a value of 0.001 to ensure stable and gradual optimization of the model parameters.

Table 1. Hyperparameters for the selected models.

Epoch	Batch Size	Learning Rate
50	32	0.001
100	32	0.001
200	32	0.001

These training parameter adjustments were made to identify the optimal combination for improving model accuracy, convergence speed, and overall generalization ability. The focus was to ensure the model could effectively detect accidents from images while minimizing overfitting and underfitting during training.

3.6. Evaluation Metrics

To assess the performance of our model in detecting vehicle accidents during adverse weather, we use key evaluation metrics such as precision (P), recall (R), accuracy, mean Average Precision (mAP), and the F1-score. These metrics provide critical insights into the model's performance, enabling us to gauge its effectiveness in detecting accidents accurately and reliably. The metrics are derived from the confusion matrix, which includes the following components:

- True Positive (TP): The model correctly detects a positive case, such as identifying a vehicle in an image with an accurate bounding box.
- True Negative (TN): The model correctly identifies the absence of an object, such as recognizing that there is no vehicle in an image.
- False Positive (FP): The model incorrectly predicts a positive case when none exists, like detecting a vehicle where there is none.
- False Negative (FN): The model fails to detect an object that is present, such as missing a vehicle that is actually in the image.

3.6.1. Precision (P)

Precision measures the accuracy of the model's positive predictions. It is calculated as the ratio of true positives (TP) to the total number of predicted positive cases (TP + FP). High precision indicates that most positive predictions made by the model are correct. It could be calculated as follows[21]:

$$P = \frac{TP}{TP + FP}$$

3.6.2. Recall (R)

Recall quantifies the model's ability to identify all relevant positive cases. It is calculated as the ratio of true positives (TP) to the total number of actual positive cases (TP + FN). High recall indicates the model's effectiveness in detecting all vehicles, even if it may result in more false positives. It could be calculated as follows[21]:

$$R = \frac{TP}{TP + FN}$$

3.6.3. Accuracy

Accuracy is a straightforward metric that measures the overall correctness of the model. It is calculated as the ratio of correctly predicted instances (both true positives and true negatives) to the total number of instances. While accuracy is useful, it may not always fully reflect performance, especially in imbalanced datasets where the number of negatives significantly outweighs positives. It could be calculated as follows [21]:

$$A = \frac{TP + TN}{TP + TN + FP + FN}$$

3.6.4. Mean Average Precision (mAP)

Mean Average Precision (mAP) is a standard performance metric in object detection. It summarizes the precision-recall curve into a single value. First, Average Precision (AP) is computed for each class by calculating precision and recall at various thresholds. mAP is then obtained by averaging the AP values across all classes, providing a holistic measure of model performance. Higher mAP values indicate better overall performance. It could be calculated as follows [22]:

$$mAP = \frac{1}{n} \sum_{i=1}^n AP_i$$

3.6.5. F1- Score

The F1-score represents the harmonic mean of precision and recall, balancing the model's accuracy in identifying positive cases and its completeness in detecting them. A high F1-score indicates a strong model performance, with both high precision and high recall, making it effective at correctly identifying accidents while minimizing both missed detections and incorrect classifications. It could be calculated as follows [26]:

$$F1 = 2 \times \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

4. Results

In this section, the results are explained after training, testing, and evaluating the models.

4.1. YOLO Object Detection model

YOLO version 10 was used for object detection of accidents in images captured during adverse weather conditions on a dataset of 1,307 images.

Table 2. Details of the results of YOLO v10

Epoch	P (%)	R (%)	mAp 50%	mAp 50-95%
50	54.2%	45.7%	44.6%	19.1%
100	63.2%	56.8%	62.9%	27.8%
200	81.8%	61%	72.5%	22.9%

YOLOv10, as shown in Table 2, achieved a Precision of 54.2%, 63.2%, and 81.8% over epochs 50, 100, and 200, respectively.

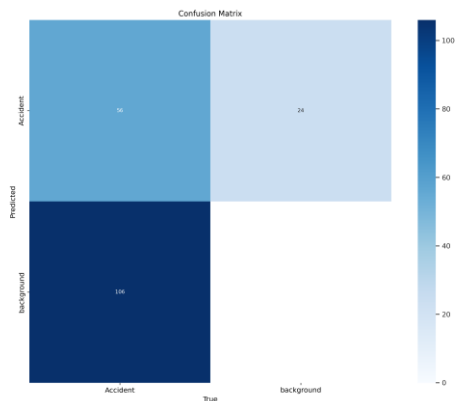


Figure 4. Confusion matrix for YOLOv10 epoch 50.

Figure 4 shows a confusion matrix for the YOLOv10 model at epoch 50, illustrating its ability to distinguish between "Accident" and "Background" classes. The matrix summarizes predictions, with diagonal values indicating accurate classifications. Higher diagonal values reflect the model's effectiveness in detecting accidents versus non-accidents at this training stage.

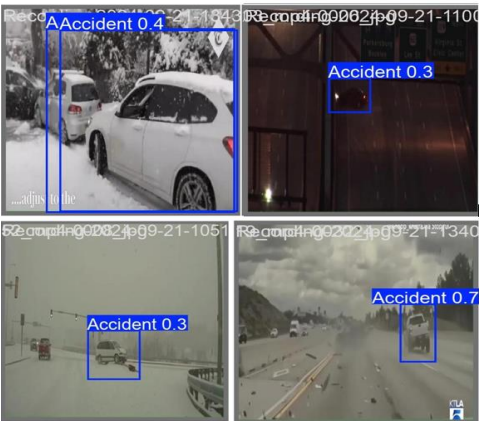


Figure 5. Train batch for YOLOv10 epoch 50.



Figure 6. Valid batch for YOLOv10 epoch 50.

Figures 5 and 6 display sample results from the training and validation batch for the YOLOv10 model at epoch 50. Detected accidents are highlighted with bounding boxes

and labeled with confidence scores, indicating the model's ability to identify and classify accident scenarios in various environmental conditions. This demonstrates YOLOv10's effectiveness in recognizing potential accidents, even in challenging visual contexts.

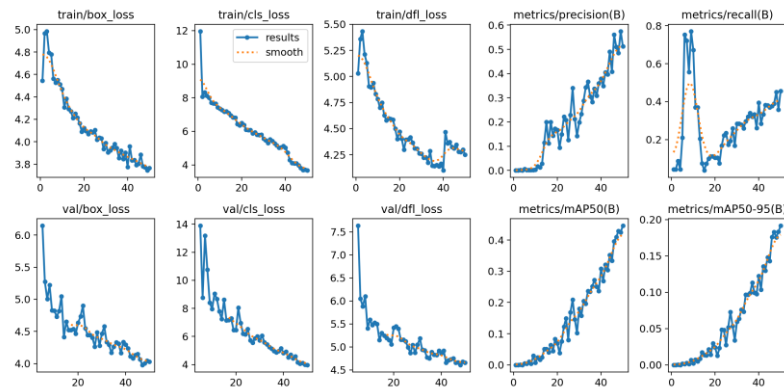


Figure 7. Training and validation loss curves epoch 50.

Figure 7 presents the training and validation loss curves, along with precision, recall, and mean average precision (mAP) metrics, for the YOLOv10 and VGG19 models over various epochs. The steady decrease in training and validation losses indicates effective learning and reduced error rates as training progresses. The upward trend in precision, recall, and mAP values demonstrates improved model performance in detecting and classifying objects. Overall, the results suggest that the models are converging well, with increasing accuracy and reliability in accident detection under adverse weather conditions.

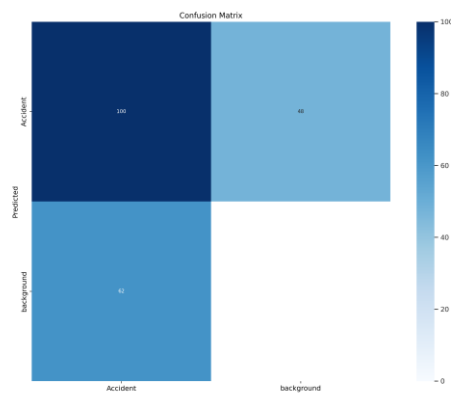


Figure 8. Confusion matrix for YOLOv10 epoch 100.

Figure 8 displays the confusion matrix for the YOLOv10 model at epoch 100. The matrix provides a breakdown of the model's classification performance between "Accident" and "Background" classes. High values along the diagonal indicate correct classifications, with a substantial number of true positives and true negatives, reflecting the model's improved accuracy at this training stage. Off-diagonal values, representing misclassifications, are relatively lower, demonstrating that the model effectively distinguishes between accident and non-accident cases. This result suggests a reliable performance for real-time accident detection.

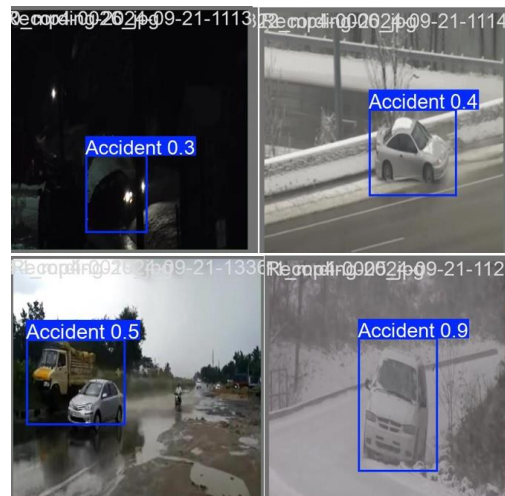


Figure 9. Train batch for YOLOv10 epoch 100



Figure 10. Valid batch for YOLOv10 epoch 100

Figures 9 and 10 display sample results from the training and validation batch for the YOLOv10 model at epoch 100. Each image shows detected accidents marked with bounding boxes and confidence scores, indicating the probability of the object being classified as an accident. The results illustrate the model's effectiveness in detecting accidents under various conditions, including low light and adverse weather, with confidence scores reflecting the model's certainty in each detection. This highlights YOLOv10's robustness and adaptability in challenging environments.

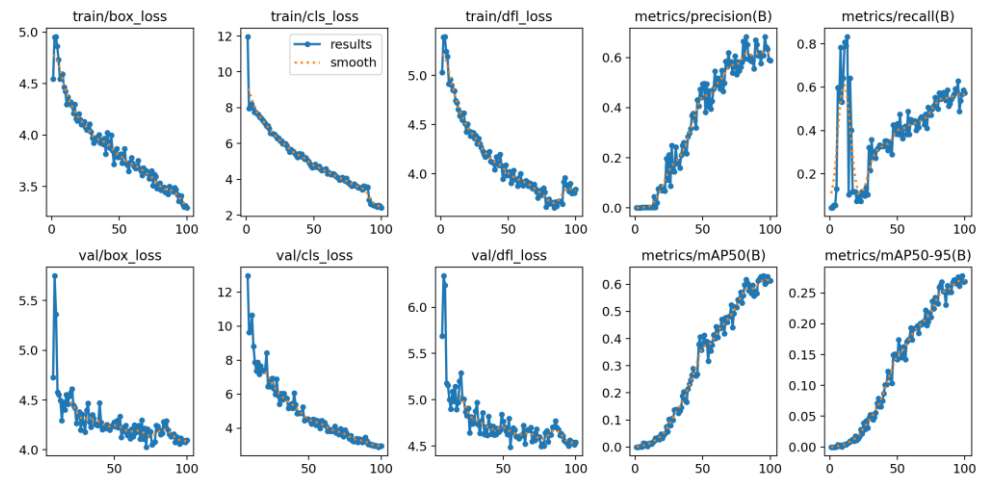


Figure 11. Training and validation loss curves epoch 100.

Figure 11 illustrates the training and validation progress of the YOLOv10 model across several metrics over 100 epochs. The box loss, classification loss, and objectness loss for both training and validation data show a consistent downward trend, indicating effective model optimization and reduced error. Meanwhile, the metrics for precision, recall, and mean average precision (mAP) display an upward trend, reflecting improved accuracy in detection performance. This progress suggests that the model is learning effectively, enhancing its capacity for accurate object detection and classification as training advances.

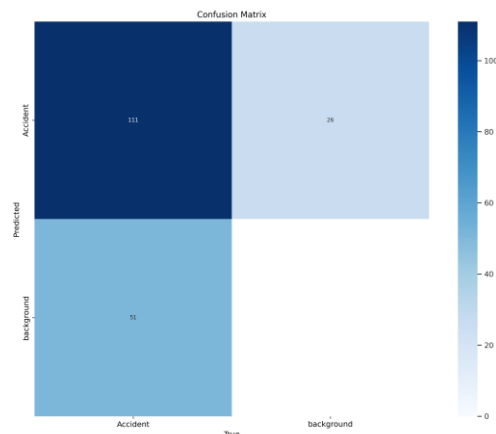


Figure 12. Confusion matrix for YOLOv10 epoch 200.

Figure 12 presents the confusion matrix for the YOLOv10 model at epoch 200. The matrix shows the model's performance in distinguishing between "Accident" and "Background" classes. High values on the diagonal represent correct classifications, indicating that the model has improved in accuracy, with a large number of true positives and true negatives. Off-diagonal values, which indicate misclassifications, are relatively low, suggesting that the model is effective in minimizing false detections. This result reflects the model's high reliability at this advanced training stage for accurate accident detection.

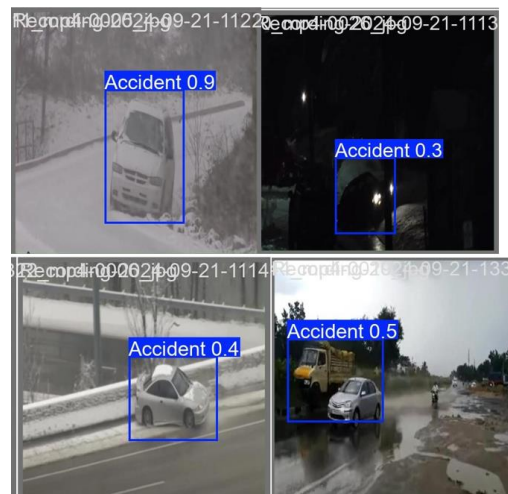


Figure 13. Train batch for YOLOv10 epoch 200.



Figure 14. Valid batch for YOLOv10 epoch 200.

Figures 13 and 14 show sample detection results from the training and validation batch for the YOLOv10 model at epoch 200. Each detected accident is highlighted with a bounding box and a confidence score, indicating the model's certainty in identifying the accident. The high confidence scores, even under varying conditions such as low light and adverse weather, illustrate the model's robustness and accuracy in accident detection at this advanced training stage. This result underscores YOLOv10's effectiveness in recognizing accident scenarios across diverse environments.

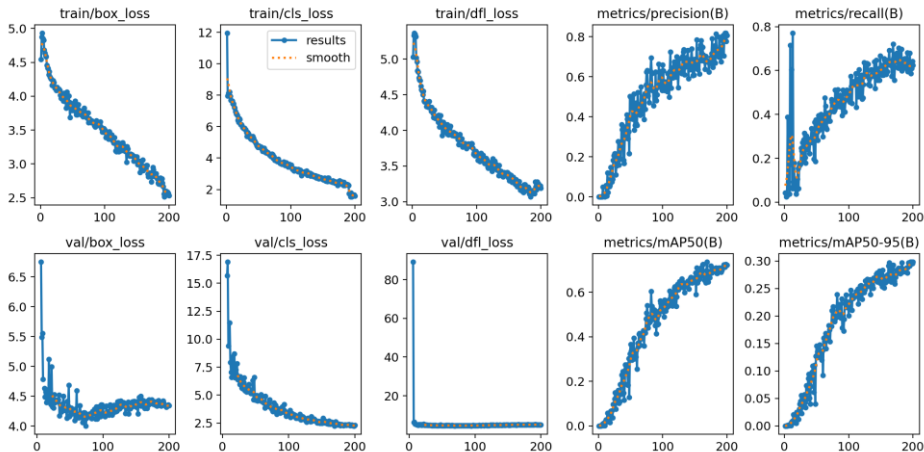


Figure 15. Training and validation loss curves epoch 200.

Figure 15 shows the training and validation performance metrics of the YOLOv10 model over 200 epochs. The box loss, classification loss, and objectness loss for both training and validation datasets consistently decrease, indicating effective learning and error reduction. Metrics for precision, recall, and mean average precision (mAP) demonstrate an upward trend, reflecting improvements in detection accuracy and consistency. The results suggest that the model is progressively learning and optimizing, achieving enhanced accuracy and reliability in identifying and classifying objects as training advances.

4.2. CNN and VGG19 models

In this part we show the comparison of two models, the CNNs with different layer configurations and VGG16 with and without a fully connected layer. The results of this analysis are shown in Table 3.

Table 3. Results of the experiments conducted on CNN and VGG19 models.

Table 3 presents the results, indicating that the VGG19 model with a fully connected layer consistently outperforms the other configurations, achieving the highest accuracy, precision, recall, and F1-score, particularly at epoch 200. This setup proves especially effective for feature extraction and classification in complex tasks. In contrast,

Model	Epoch	Learning Rate	Batch Size	Train Loss	Train Acc.	Valid Loss	Valid Acc.	Precision	Recall	F1-Score
CNN 7 layers without dropout	50	0.001	32	0.6932	0.4743	0.9699	0.4970	0.25	0.50	0.33
	100	0.001	32	0.6931	0.5176	0.7046	0.4940	0.25	0.50	0.33
	200	0.001	32	0.6932	0.4743	0.9505	0.4970	0.25	0.50	0.33
CNN 7 layers with dropout	50	0.001	32	0.660	0.6195	0.7160	0.5030	0.25	0.50	0.32
	100	0.001	32	0.6757	0.5976	0.7249	0.5029	0.26	0.51	0.34
	200	0.001	32	0.6792	0.5900	0.7431	0.5029	0.27	0.50	0.34
CNN 10 layers without dropout	50	0.001	32	0.6563	0.6375	0.7026	0.5357	0.29	0.54	0.37
	100	0.001	32	0.6661	0.6166	0.7054	0.5357	0.29	0.54	0.37
	200	0.001	32	0.6633	0.6216	0.7072	0.5357	0.29	0.54	0.37
VGG19 with fully connected layer	50	0.001	32	0.1758	0.9156	0.4792	0.8452	0.8457	0.8453	0.8457
	100	0.001	32	0.1080	0.9548	0.6222	0.8333	0.8395	0.8333	0.8335
	200	0.001	32	0.1459	0.9219	0.6982	0.8512	0.8531	0.8512	0.8514
VGG19 without fully connected layer	50	0.001	32	0.6527	0.6465	0.7024	0.5357	0.29	0.54	0.37
	100	0.001	32	0.6717	0.6136	0.8017	0.5357	0.29	0.54	0.37
	200	0.001	32	0.6658	0.6176	0.7429	0.5357	0.29	0.54	0.37

CNN models with and without dropout demonstrate moderate performance, with dropout slightly enhancing recall and F1-score, which suggests improved generalization. The 10-layer CNN without dropout shows minimal improvements in accuracy, as the absence of dropout limits its effectiveness despite the increased depth. Overall, the VGG19 model with a fully connected layer emerges as the most reliable and accurate configuration among those evaluated.

5. Discussion

The results of this study underscore the effectiveness of a dual-model deep learning approach for vehicle accident detection in adverse weather conditions. By combining YOLOv10's real-time object detection capabilities with VGG19's detailed feature extraction, the hybrid framework achieved a significant increase in detection accuracy and a reduction in false positives compared to single-model approaches. These findings align with prior research that highlights the limitations of conventional single-model frameworks in accurately detecting accidents under challenging visibility conditions. In contrast to simpler detection frameworks, which often struggle in low-visibility scenarios like fog, rain, and snow, our hybrid approach leverages both models' unique strengths. YOLOv10's rapid object detection ability enables it to quickly identify potential accident-related objects such as vehicles, road elements, and pedestrians, even under adverse conditions. VGG19's hierarchical structure allows for intricate feature extraction from

these identified regions, enhancing the system's ability to recognize subtle indicators of accidents, such as vehicle damage or unusual positioning. The combination of these models is further strengthened by adaptive preprocessing techniques, including dehazing and contrast adjustments, which improved image clarity and enabled more accurate analysis by both YOLOv10 and VGG19. This dual-model approach demonstrated an accuracy rate of 81.8% at 200 epochs, indicating that integrating multiple deep learning models with preprocessing techniques can yield robust accident detection even in visually complex environments. The study's findings are consistent with prior works, such as those by Zhou et al. and Chen et al., which showed that combining deep learning models with adaptive preprocessing techniques can enhance detection performance in adverse conditions. However, this study takes a unique approach by using real-time detection capabilities of YOLOv10 in tandem with the analytical depth of VGG19, specifically for accident detection, rather than general object detection. This focused application of a hybrid model significantly advances traffic safety research by providing a reliable tool for accident detection in environments where rapid emergency response is critical. While this research provides a solid foundation for real-time accident detection, several areas for improvement remain. Future studies may consider integrating additional sensor data, such as LiDAR and radar, to further improve detection accuracy, especially in cases where environmental noise or extreme conditions limit visual data reliability. Additionally, the use of synthetic data for training could be expanded to simulate a wider range of weather conditions, enhancing the models' generalizability across diverse traffic scenarios. Another promising avenue is exploring more advanced versions of YOLO or other real-time models to further boost detection speed without compromising accuracy. This study presents a novel, hybrid deep learning approach for accident detection that addresses the unique challenges posed by adverse weather conditions. The integration of YOLOv10 and VGG19, supported by adaptive preprocessing techniques, marks a significant advancement in accident detection research, contributing to the broader goal of intelligent transportation systems, supporting improved road safety, and highlighting the potential of multi-model frameworks for critical applications requiring high accuracy and responsiveness. Future research focused on expanding the dataset, incorporating additional environmental factors, and refining model architectures could further enhance the robustness and adaptability of accident detection systems, supporting safer and more efficient road networks global.

6. Conclusion

This study introduced a dual-model framework for vehicle accident detection under adverse weather conditions, combining YOLOv10 for real-time object detection with VGG19 for detailed feature extraction. This approach addressed limitations of traditional methods by enhancing detection accuracy and reducing false positives through the complementary strengths of both models. Preprocessing techniques like dehazing and contrast adjustment further improved image clarity, achieving an accuracy rate of 81.8% at 200 epochs.

Future work may focus on integrating additional sensors (e.g., LiDAR) and expanding synthetic data for greater adaptability across diverse weather scenarios. This framework provides a practical solution for accident detection in challenging environments, supporting safer and more responsive road systems.

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